TABLE 1 Efficiency of homologous recombination

(*a*) Disruption of *pim-1* with *neo*

DNA	Number of cells E14 pim^+/pim^+	$G418^R$	Homol. rec.
39	10^8	1.5×10^4	0/4
31	10^8	8×10^2	34/40 (85%)

(*b*) Disruption of *pim-1* with *hyg*

DNA	Number of cells E14 pim^+/pim^+	Hyg^R	Homol. rec.
pPGK *Hyg*	2×10^7	700	
61 A^+	2×10^7	235	N.D.
61 A^-	2×10^7	115	N.D.
59 A^-	2×10^7	13	6/11 (55%)
59 A^-	10^8	48	20/31 (64%)

(*c*) Disruption of both *pim-1* alleles with *neo* and *hyg*

DNA	Number of cells E14 pim^+/neo^+	Hyg^R	Homol. rec.
59 A^-	10^8	43	21/34 (62%)*

Efficiency of homologous recombination with *pim-1* targeting constructs. ES cell line E14 was grown on BRL conditioned medium as described[10]. Cells (10^8 or 2×10^7) were mixed with 150 or 100 μg of DNA, respectively, in a volume of 600 μl PBS buffer and electroporated using a Biorad Gene pulser (0.8 kV, 3 μF, electrode distance 0.4 cm). Cells were reseeded on 10-cm tissue culture dishes at a density of 10^7 cells per plate. G418 selection (200 μg ml^{-1}) was started after one day and colonies were picked and expanded after 6 days; hygromycin B selection (110 μg ml^{-1}) was started after 2 days and colonies were picked after 12 days.

a, Electroporation of E14 ES cells with the *neo* constructs 39 (control) and 31 (targeting construct).

b, Electroporation of E14 ES cells with the *hyg* constructs 61 (control) plus and minus polyadenylation signal and $59A^-$ (targeting construct). A positive control for hygromycin B selection in ES cells was constructed as the Pim–Hyg fusion protein may have a reduced activity. This construct, pPGK-*hyg*, contains the PGK promoter[11], which is highly active in ES cells, an optimized translation initiation site[12], the coding sequence for the bacterial *hyg* gene, and the PGK polyadenylation signal. The number of ES cell colonies resistant to hygromycin B was markedly lower with $61A^+$ than with pPGK *hyg* (*b*). Only a twofold reduction in colony formation was seen if the *pim-1* polyadenylation signal was omitted from the *pim–hyg* fusion construct (compare $61A^+$ and $61A^-$). The absence of a promoter and a translational start codon further reduced the number of colonies resistant to hygromycin B tenfold ($59A^-$). Fifty five per cent of these colonies (6 out of 11 tested) were the result of homologous recombination. In a separate experiment at a larger scale, 64% of the colonies resistant to hygromycin B (20 out of 31 tested) contained the *hyg* sequence integrated in one of the *pim-1* alleles. N.D., not determined.

c, Electroporation of one of the ES clones previously targeted with *pim–neo* construct 31 with the *pim–hyg* targeting construct $59A^-$.

* $G418^SHyg^R$: 6 → single knock-out. $G418^RHyg^R$: 15 → double knock-out.

differentiate *in vitro* on agarose-coated plates to which they cannot adhere. Characteristic embryoid bodies, containing contractile cells, developed in all cases. When reseeded on tissue culture plates, a variety of different cell types grew out from the embryoid body structures. No microscopically discernable differences were observed at this stage between cells with or without a functional *pim-1* gene. This suggests that in spite of its high expression in ES cells, *pim-1* is not required for normal ES cell proliferation and differentiation *in vitro*. Its function in ES cells is therefore unclear at present. As the Pim-1 serine/threonine protein kinase is thought to be involved in signal transduction, we might assume that differences in growth behaviour or differentiation potential might nevertheless become apparent by exposing the cells to different growth conditions and growth factors.

In conclusion, our results show that complete loss-of-function mutations in expressed genes can indeed be directly generated in diploid ES cells with high efficiency using targeting vectors containing promoter/AUG-minus *neo* and *hyg* fusion genes. In our experiment the selection procedure used to inactivate the *pim-1* gene was particularly efficient. But data from other laboratories using different selection protocols indicate that a large variety of genes[9], expressed and nonexpressed, can be targeted with a sufficiently high efficiency. Selection protocols employing a promoter-competent *neo* gene as a positive selection marker[1] should be readily modifiable for inactivation of the residual allele by substituting *hyg* or *neo*. In cases in which the gene of interest is expressed in cultured cells, there is now the option to study the effects of loss-of-function mutations directly *in vitro*, without the necessity to generate germ-line chimaeric mice as an intermediate step. Besides this application *in vitro*, chimaeric mice produced with ES cells lacking a particular gene function might reveal somatic abnormalities that are not accessible to analysis by crossing of heterozygous mutants because of early lethality. □

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A mutation in the $tRNA^{Leu(UUR)}$ gene associated with the MELAS subgroup of mitochondrial encephalomyopathies

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MITOCHONDRIAL encephalomyopathies are usually divided into three distinct clinical subgroups: (1) mitochondrial myopathy, encephalopathy, lactic acidosis and stroke-like episodes (MELAS); (2) myoclonus epilepsy associated with ragged-red fibres (MERRF); and (3) chronic progressive external ophthalmoplegia (CPEO) including Kearns–Sayre syndrome[1–5]. Large deletions of human mitochondrial DNA and a transition mutation at the mitochondrial transfer RNA^{Lys} gene give rise to CPEO including Kearns–Sayre syndrome[6–8] and MERRF[9,10], respectively. Here we report an A-to-G transition mutation at nucleotide pair 3,243 in the dihydrouridine loop of mitochondrial $tRNA^{Leu(UUR)}$ that is specific to patients with MELAS. Because this mutation creates

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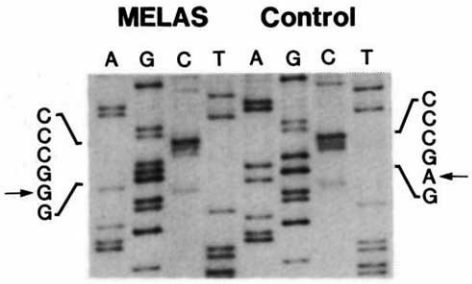

FIG. 1 Direct sequencing of the flanking region of nucleotide pair (np) 3,243 in a MELAS patient and a control. Nucleotide A in wild type is converted to a G at position 3,243. Because 22 tRNA genes are clustered in 11 locations, we synthesized 11 sets of 20-mer oligonucleotides with PCR-MATE (Applied Biosystems) as follows: np 520–539 (forward: f) and np 726–707 (reverse: r) for $tRNA^{Phe}$; np 1,549–1,568 (f) and np 1,722–1,703 (r) for $tRNA^{Val}$; np 3,130–3,149 (f) and np 3,423–3,404 (r) for $tRNA^{Leu(UUR)}$; np 4,101–4,120 (f) and np 4,531–4,512 (r) for $tRNA^{Ile}$, $tRNA^{Gln}$ and $tRNA^{Met}$; np 5,470–5,489 (f) and np 5,928–5,909 (r) for $tRNA^{Trp}$, $tRNA^{Ala}$, $tRNA^{Asn}$, $tRNA^{Cys}$ and $tRNA^{Tyr}$; np 7,375–7,394 (f) and np 7,914–7,895 (r) for $tRNA^{Ser(UCN)}$ and $tRNA^{Asp}$; np 8,211–8,230 (f) and np 8,410–8,391 (r) for $tRNA^{Lys}$; np 9,912–9,931 (f) and np 10,555–10,536 (r) for $tRNA^{Gly}$ and $tRNA^{Arg}$; np 11,798–11,817 (f) and np 12,519–12,500 (r) for $tRNA^{His}$, $tRNA^{Ser(AGY)}$ and $tRNA^{Leu(CUN)}$; np 14,553–14,572 (f) and np 15,230–15,211 (r) for $tRNA^{Glu}$; and np 15,806–15,825 (f) and np 16,219–16,190 (r) for $tRNA^{Thr}$ and $tRNA^{Pro}$. The double-stranded DNA fragments were amplified by the conventional PCR procedure using each set of primers[26]. After electrophoretic separation, reamplification of an aliquot of elute from a selected gel slice with unidirectional or, if necessary, bidirectional asymmetric PCR was carried out to produce single-stranded templates[27]. They were purified with a microconcentrator (Centricon 30, Amicon), and sequenced by the dideoxy chain termination method with Sequenase Version 2.0 (USB)[28].

an *Apa*I restriction site, we could perform a simple molecular diagnostic test for the disease. The mutation was present in 26 out of 31 independent MELAS patients and 1 out of 29 CPEO patients, but absent in the 5 MERRF and 50 controls tested. Southern blot analysis confirmed that the mutant DNA always coexists with the wild-type DNA (heteroplasmy).

Recent studies on the expression of defective mitochondrial (mt) DNA suggest that tRNA dysfunction is the cause of CPEO and MERRF[9–13]. It is reasonable to assume that there are also tRNA abnormalities in MELAS, because all three mitochondrial encephalomyopathies are histologically similar and often have a defect(s) of respiratory-chain enzymes, in particular NADH coenzyme Q reductase and cytochrome *c* oxidase. There is no one-to-one correspondence between enzyme deficiencies and clinical entities, suggesting that mutations at tRNA or ribosomal RNA genes are more responsible for MELAS than are mutations of protein-coding regions. Therefore, we sequenced all mitochondrial tRNA genes of a MELAS patient from a pedigree in which the disease is maternally inherited. We directly sequenced the amplified DNA using the polymerase chain reaction (PCR) with 11 sets of 20-residue oligonucleotide primers (Fig. 1). Compared with reference sequences[14], only one nucleotide substitution was detected, occurring at nucleotide pair 3,243 in the coding region of the $tRNA^{Leu(UUR)}$ gene. This corresponds to the first base of the dihydrouridine loop and is strictly invariant from humans to sea urchins[14–23] (Fig. 2). This base also forms a tertiary hydrogen bond with the T residue of the amino-acid acceptor arm positioned around a turn of L-shape structure[24]. Hence, any substitution at this position can greatly affect $tRNA^{Leu(UUR)}$ function.

TABLE 1 Nucleotides A or G at position 3,243

	Total	A	G
MELAS	31	5	26
MERRF	5	5	0
CPEO	29	28	1
Control	50	50	0

This mutation allowed us to develop a simple molecular diagnostic test for the disease which uses the restriction endonuclease *Apa*I. This enzyme can cleave the mutant sequence (GGGCCC) but not the wild-type sequence (GAGCCC). A 294-base-pair (bp) fragment encompassing the mutation site was therefore amplified by PCR and digested with *Apa*I. Each set of PCR fragments from 26 of 31 independent MELAS patients was cleaved to give three fragments (the original 294-bp fragment and the new 182-bp and 112-bp fragments), but those from 50 normal controls remained as copies of the 294-bp fragment even after overnight digestion. Thus the mutation at nucleotide pair 3,243 seems to be most responsible for MELAS. Two other members in the pedigree, including the original patient, have the mutation. Five patients without the mutation, however, show all the clinical and histological features of MELAS, indicating that there is another molecular cause of MELAS. An examination of the digestion profile always shows the coexistence of mutant and wild-type genomes (heteroplasmy). This was confirmed by Southern blotting:populations of mutant DNA ranged from 50 to 92% using densitometry (Fig. 3).

To determine whether this mutation is specific to MELAS, we examined samples from other mitochondrial diseases (Table 1). All of the five MERRF patients examined had a point mutation at nucleotide pair 8,344 but lacked the mutation at nucleotide pair 3,243. Out of 29 CPEO patients examined, 28 lacked the mutation. Of these 28, 25 had single large deletions, two had multiple deletions and one had no deletions, as observed on Southern blot analyses. The CPEO patient who carried the mutation had no large deletion of mtDNA. His mother and sister also showed clinical features of mitochondrial diseases, having CPEO and muscle weakness; the sister had periodic central nervous system symptoms with focal low lucency (shown

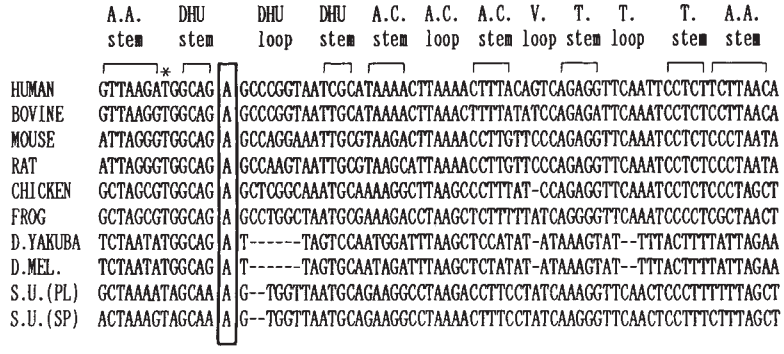

FIG. 2 Interspecies similarity of the mitochondrial $tRNA^{Leu(UUR)}$ genes. An adenine, or the first base of the dihydrouridine (DHU) loop in $tRNA^{Leu(UUR)}$, is boxed and is evolutionarily conserved among human[14], bovine[15], mouse[16], rat[17], chicken[18], frog[19], two species of fly[20,21] and two species of sea urchin[22,23]. The base forms a hydrogen bond to T(*), (U in the tRNA) to construct a functional tertiary L-shape structure, so the substitution probably causes the dysfunction leading to suppression of protein synthesis. A.A., amino acid; A.C., anticodon; V., variable; T., TψC loop; D. YAKUBA, *Drosophila yakuba*; D. MEL., = *Drosophila melanogaster*; and S.U., sea urchin (PL, *Paracentrotus lividus*; SP, *Strongylocentrotus purpuratus*).

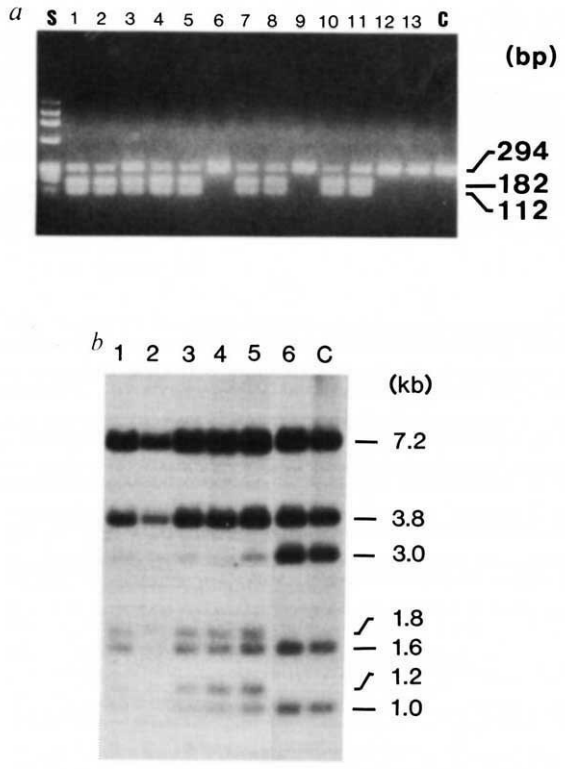

FIG. 3 The digestion profiles of PCR fragments amplified with a set of primers for $tRNA^{Leu(UUR)}$ and Southern blot analyses. *a*, Point mutation (A to G) at nucleotide pair 3,243 was found in MELAS patients but not in controls. Size-standard DNA (S) was ΦX174 DNA digested by *Hae*III. Fragments of 294 bp amplified by conventional PCR using a set of primers for $tRNA^{Leu(UUR)}$ (Fig. 1) were digested with the restriction endonuclease *Apa*I for more than 6 h. With samples from muscles of MELAS patients (lanes 1–5, 7, 8, 10, 11), an original and two new fragments (182 bp and 112 bp) could be recognized on electrophoresis after staining with ethidium bromide. By contrast, fragments of 294 bp from some MELAS patients (lanes, 6, 9, 12, 13) and that amplified with the placental DNA of a control as template (lane C) remained unchanged after the digestion with *Apa*I. *b*, Southern blot analyses of total DNAs from patients and a control after *Apa*I digestion. In most MELAS patients (lanes 1–5), varying amounts of 3.0-kilobase (kb) fragments were cleaved into two smaller ones of 1.8 and 1.2 kb because of the site-gain mutation at nucleotide pair 3,243, and the rest of the 3.0-kb fragments remained unchanged, indicating the presence of wild-type DNA (heteroplasmy). One MELAS patient (lane 6) and one control (lane C) show no 1.8- and 1.2-kb bands after *Apa*I digestion.

by tomography) similar to that of MELAS[25]. Although we could not evaluate the importance of the site-gain mutation in the male CPEO patient, it is likely that the mutant mtDNA is not abundant enough for the clinical symptoms to be manifest. Alternatively, the effect of the abnormalities leading to CPEO might be much more severe than that of the point mutation leading to MELAS. We conclude that the mutation at nucleotide pair 3,243 is specific, although not exclusive, to MELAS.

How abnormalities of tRNAs cause functional and morphological changes of mitochondria, and why types of tRNA mutations are generally correlated with types of mitochondrial diseases, remain to be answered. Other factors, particularly nuclear factors, could be involved in the pathogenesis of mitochondrial diseases. Nonetheless, the mutation at nucleotide pair 3,243 seems to be the main cause of MELAS, and all three distinct subgroups of mitochondrial diseases may share similar mitochondrial tRNA abnormalities. □

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Evolution of a polymeric globin in the brine shrimp *Artemia*

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SEVERAL invertebrate species possess haemoglobins in which each polypeptide contains multiple haem-binding domains[1], possibly reflecting the fusion of multiple monomeric globin transcriptional units at the gene level. We have now analysed the transcript of such a polymeric globin gene from the brine shrimp *Artemia*, which expresses three polymeric haemoglobins, each of relative molecular mass 260,000 (M_r 260K). These are formed by the variable association of two different subunit types, α and β (refs 2, 3). Haemoglobins I and III are homodimers of α and β subunit types, respectively, and haemoglobin II is a heterodimer ($\alpha\beta$). The individual globin chains are of similar size (M_r 130K), but the exact nature of the differences between the two subunit types is unclear. Analysis of complementary DNA clones encoding one of the subunits of the *Artemia* dimeric haemoglobin showed that the globin messenger RNA encodes nine myoglobin-like domains, connected by linking peptides. The residues in the linkers are characteristic of those found generally in such protein linkers, and include turn-promoting amino acids. Each domain also contains the conserved residues that are required for functional haem-binding, and from analysis of the sequences it was predicted that they all can adopt the classic myoglobin-like fold. Analysis of the derived amino-acid sequences indicated that the individual domains are duplicated monomers that fused to form the polymeric globin some 200 Myr ago. The fusion of multiple transcriptional units for the evolution of a polymeric globin gene may have been a general mechanism for the appearance of such polymeric haemoglobins in invertebrates.

We isolated several *Artemia* globin complementary DNA clones[4] and determined the complete DNA and derived amino-acid sequence of the largest (Fig. 1). The authenticity of the